



Design and performance analysis of joint control and shared channel scheduler for downlink in 3GPP narrowband-IoT

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ABSTRACT

Narrowband-IoT (NB-IoT) is a low power wide area network technology introduced by the 3rd Generation Partnership Project (3GPP) in Release 13 specifications. The NB-IoT aims at providing cellular connectivity to low cost, low throughput, delay-tolerant devices, and hence, is a key technology for smart connected living. The NB-IoT specifications continue to evolve as part of 5G New Radio (NR), and the technology is expected to co-exist with 5G NR. A transmission bandwidth of 180 KHz is required to deploy the NB-IoT. In NB-IoT, the duration of the physical downlink control channel (NPDCCH) and the physical downlink shared channel (NPDSCH) changes dynamically. When the base station broadcasts the NPDCCH region, all the active NB-IoT devices try to decode it. Hence, to reduce the power consumption of the NB-IoT devices, more number of NB-IoT devices have to be scheduled in each scheduling attempt. For each payload in NPDCCH, there is a respective payload in NPDSCH. Thus, a base station should schedule the devices simultaneously in both NPDCCH and NPDSCH to reduce resource wastage. The time-frequency resources in NB-IoT are limited and valuable. Further, the NB-IoT devices have low battery power constraint. Thus, scheduling the devices by considering the above constraints has a significant impact on the NB-IoT system performance. Motivated by this, we propose a joint scheduling algorithm for both NPDCCH and NPDSCH. Further, we propose an adaptive repetition allocation algorithm and present a receiver design for NB-IoT. We also define performance metrics to evaluate the impact of the proposed scheduler and then present bounds on the performance gains with the proposed scheduler. With system-level and link-level simulations, we show that the proposed scheduler significantly outperforms the current state-of-the-art algorithms.

1. Introduction

The Narrowband Internet of Things (NB-IoT) is a low power wide area network technology introduced by the 3rd Generation Partnership Project (3GPP) in Release 13 [1–4]. The NB-IoT devices are low cost, low throughput, and delay-tolerant. The technology aims at providing cellular connectivity for various use cases like smart metering, object tracking, and connecting industrial appliances. The NB-IoT specifications continue to evolve as part of 3GPP 5G New Radio (NR), and the technology is expected to co-exist with 5G NR. The NB-IoT has the physical layer design similar to that of Long Term Evolution (LTE). It requires a transmission bandwidth of 180 KHz and can operate in three modes, namely Standalone, Guardband, and In-band mode. In Standalone mode, NB-IoT operates at any carrier frequency with 180 KHz bandwidth. In Guardband mode, NB-IoT operates in the resources available in the guard band of LTE. For In-band mode, NB-IoT co-exists with LTE/5G-NR and operates in one single physical resource block of LTE/5G-NR [4,5].

The NB-IoT has three physical channels in the downlink, namely physical broadcast channel (NPBCH) for transmitting the base station related information, physical downlink control channel (NPDCCH) for conveying the downlink/uplink scheduling grants, and physical downlink shared channel (NPDSCH) to carry the data intended for the NB-IoT device. Downlink control information (DCI) is the payload transmitted in NPDCCH. The DCI conveys all the information required to decode the data transmitted in NPDSCH. However, the NB-IoT device does not know the location of DCI, and hence, it does blind decoding in the NPDCCH region for the DCI. To reduce these blind decoding attempts at the device, 3GPP specifies that a DCI can be scheduled only at prescribed subframes inside NPDCCH. Further, the NB-IoT devices can operate in poor signal coverage, and hence, the data is repeated over many subframes in NPDCCH and NPDSCH for a successful reception. With each device requiring different repetitions and with the previously mentioned 3GPP constraints, there are possibly many ways to allocate resources to devices in the downlink.

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The time–frequency resources in the NB-IoT are limited. For the In-band mode, the NB-IoT operates in the resources of LTE/5G-NR, and hence, the time–frequency resources are valuable. The NB-IoT devices require long-lasting battery power, and hence, the devices should consume less power. Thus, resource utilization and power consumption of the devices are the key performance indicators of the NB-IoT system. The set of subframes constituting NPDCCH and NPDSCH is broadcast to the devices by the base station and is called an NPDCCH period (NP). As per 3GPP specifications [3], if the base station schedules DCI of an NB-IoT device in NPDCCH of an NP, then the NPDSCH of the same NP should carry the data for the corresponding device. Hence, to reduce the resource wastage, there is a need for allocating resources to the devices by simultaneously considering both NPDCCH and NPDSCH requirements. Further, when the base station broadcasts the NP, all the active NB-IoT devices in the system try to decode the NP. As per 3GPP constraints [3], an NP can accommodate a maximum of eight NB-IoT devices. Thus, more the number of devices scheduled in an NP, less is the overall power consumption of the devices. The joint scheduling of NB-IoT devices in NPDCCH and NPDSCH, considering the 3GPP constraints is a challenging problem and can improve the resource utilization and power consumption of the devices. In the direction, the joint scheduling problem for NB-IoT has been considered in [6]. However, the adaptation of repetition allocation for the NB-IoT devices has not been considered in [6]. Further, the essential aspect of receiver design is also missing in [6]. Motivated by this, the main contributions of this work (as an extension of [6]) are as follows.

1. We propose an adaptive repetition allocation algorithm and present a receiver design for the downlink in NB-IoT.
2. We define performance metrics to evaluate the proposed scheduler and present the bounds on the performance gains with the proposed scheduler.
3. We perform system-level and link-level simulations, and show that the proposed algorithm significantly outperforms the state-of-the-art algorithms.

The rest of the paper is organized as follows. In Section 2, we present the related work in the literature. Section 3 describes the scheduling procedure in NB-IoT as per 3GPP constraints. In Section 4, we define performance metrics, propose the joint scheduling algorithm, repetition algorithm, and receiver design. In Section 5, we present the simulation results. Section 6 discusses some concluding remarks and possible future works.

2. Related work

In [7], joint resource and power allocation for control channel of 3GPP-LTE has been discussed. The number of resources to be allocated for a user is varied with the power allocated to that user. This way, irrespective of the user density, the power and resource optimization is achieved at the base station. Further, in [7,8], simple sorting based algorithms have been presented for scheduling the users in the control channel of LTE. In [9,10], authors have presented a novel rate matching scheme for the downlink control channel, which reduces the power consumption of the devices in LTE-M1 technology. In [11], optimal resource allocation of the control channel has been performed for 3GPP-LTE. To achieve this, authors have formulated resource allocation as a set-packing problem. With Monte-Carlo simulations, authors have shown that the optimization significantly increases the system's cell-edge performance. In [12], various scheduling algorithms have been discussed in the context of LTE. The authors have considered guaranteed bit rate, constraints from random access-procedure, and real-time traffic models while evaluating the proposed scheduler designs. Through system-level simulation results, authors have shown that the proposed algorithms improve the QoS for real-time traffic. In [13], various scheduling procedures and respective trade-offs between energy efficiency, spectral efficiency, and fairness have been analysed for

the cloud radio access network architectures. In [14], authors have presented low complexity scheduling algorithms for OFDMA systems with guaranteed fairness. In [15], authors have proposed a new remapping strategy based on a linear transformation matrix for the LTE PDCCH. Through simulation results, authors have shown that the proposed scheme has better resource utilization and less probability of blocking. In [16], authors have presented a scheduling algorithm based on the radio network temporary identifier of the user. Further, authors have shown that the system performance improves significantly in terms of resource utilization, allocation delay time, and probability of blocking. However, all these algorithms are in the context of 3GPP-LTE, and cannot be directly applied to NB-IoT.

In [17], the constraints and the scheduling process for the downlink physical layer of 3GPP NB-IoT have been discussed. The problem of delay assignment for the optimal UE scheduling has been explained and some heuristic solutions have been provided. However, the issue of deciding the length of NP has not been addressed. In [18], an optimal scheduling algorithm for the NPDSCH has been proposed. However, the authors have assumed that the information of the NPDCCH region and the NB-IoT devices to be scheduled in each NPDCCH region are available a priori. The problem of defining the NPDCCH region has not been addressed in [18]. In [19], an optimal control channel scheduling algorithm has been presented for NPDCCH. The scheduling problem of the control channel is formulated with 3GPP constraints and is optimized for better resource utilization and power consumption of the NB-IoT devices. However, none of these existing works address the issue of joint scheduling of both control and data channel regions. [6] is the first attempt to consider the resource requirements of each NB-IoT device in both control and shared channel while scheduling the devices. Further, in this paper, we calculate the optimal length of the NP, and then, solve an optimization to schedule the devices in NPDCCH and NPDSCH of the NP. For performance comparison, we have implemented the scheduling algorithms presented in [19]. Next, we present the scheduling in NB-IoT and the constraints as per 3GPP specifications. Further, for ease of reading, a list of acronyms used in this paper is presented in Table 1.

3. Scheduling in NB-IoT

Whenever an NB-IoT device is switched on, it initially performs the cell search and synchronization procedure to get connected to the base station. For this synchronization, the device uses NB-IoT primary and secondary synchronization signals termed as NPSS and NSSS, respectively. As shown in Fig. 1, the base station transmits NPSS in every 6th subframe of each radio frame and NSSS in 10th subframe of every alternate radio frame. In case the NB-IoT device is successful in the synchronization procedure, it starts decoding the NPBSCH. The base station transmits NPBSCH on SF 0 of every radio frame, as shown in Fig. 1. Then, for the NB-IoT device to communicate with the base station or vice-versa, the device needs to know the region over which it can transmit or receive the data. For this purpose, the base station transmits DCI in the NPDCCH. The DCI carries all the information required by the NB-IoT device to decode the data in NPDSCH. It carries the information about the exact location of the downlink grants, modulation and coding scheme, and downlink data duration. All the subframes excluding the NPSS, NSSS, and NPBSCH are available for NPDCCH/NPDSCH, as shown in Fig. 1. However, the base station conveys the duration of NPDCCH and NPDSCH to the NB-IoT devices. The base station announces only the NPDCCH region, and thus, the exact location of the DCI is not known a priori to the NB-IoT device. In the NPDCCH region, the NB-IoT device performs a blind search for the DCI. However, 3GPP constraints limit the number of such blind decodings. In case the DCI is successfully decoded, the NB-IoT device then decodes the actual data intended for it in the NPDSCH. Next, we explain the scheduling of the devices in NPDCCH in detail.

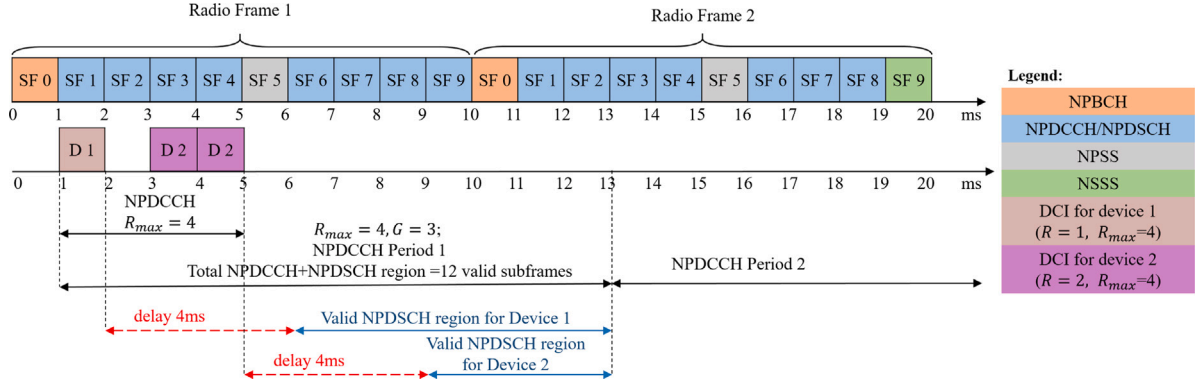


Fig. 1. Scheduling of NPDCCH and NPDSCH in NB-IoT.

Table 1

List of Acronyms.

3GPP	3rd Generation Partnership Project
α	Tunable parameter to achieve various trade-offs
β	Threshold for block-error rate
η	Number of devices scheduled per NP
χ	Resource utilization
BLER	Block Error Rate
CCes	Control Channel Elements
CDF	Cumulative Distribution Function
CRC	Cyclic Redundancy Check
CP	Cyclic Prefix
CSS	Common Search Space
DCI	Downlink Control Information
G	Parameter to define the NP duration
IFFT	Inverse Fast Fourier Transform
k_0	Delay between control and shared channel
LTE	Long Term Evolution
MCS	Modulation and Coding Scheme
N_{rep}	Repetition required for the shared channel data
N_{SF}	Subframes of data required by the device
NB-IoT	Narrowband Internet of Things
NP	NPDCCH Period
NPBCH	Narrowband Physical Broadcast Channel
NPDCCH	Narrowband Physical Downlink Control Channel
NPDSCH	Narrowband Physical Downlink Shared Channel
NR	New Radio
OFDM	Orthogonal Frequency Division Multiplexing
OLLA	Outer-Loop Link Adaptation
PRB	Physical Resource Block
RE	Resource Element
RNTI	Radio Network Temporary Identifier
RRC	Radio Resource Control
R	Repetition required for the control channel data
R_{max}	Number of subframes in a search space
SINR	Signal-to-Interference plus Noise Ratio
TB	Transport Block
USS	UE Specific Search Space

3.1. NPDCCH scheduling

In NB-IoT, a resource element (RE) is the smallest time–frequency resource consisting of one subcarrier during one symbol. A physical resource block (PRB) comprises of 14 OFDM symbols over 12 subcarriers and consists of 168 REs. The NB-IoT operates with 180 kHz of transmission bandwidth, and thus, has one PRB in 1 ms. The REs available in one PRB form two control channel elements (CCEs). In NPDCCH, the DCIs of the NB-IoT devices get scheduled in units of CCEs [1]. Further, to ensure the successful reception to a device in the poor coverage, the scheduled DCI may repeat over (R) subframes. The number of CCEs and R for each device are decided based on the channel conditions of the NB-IoT device. When the base station announces the NPDCCH region over a certain number of subframes (R_{max}) [3], all the active NB-IoT devices start decoding the region for the DCI. Since the

NB-IoT devices do not know the exact location of the DCI, they blindly decode the entire region. To reduce the blind decoding attempts by the NB-IoT device, 3GPP specifies certain constraints [3] on the scheduling of the devices in NPDCCH as shown in (1)–(2).

$$R_{max} \in \{1, 2, \dots, 2048\}. \quad (1)$$

$$R \in \left\{ R_{max}, \frac{R_{max}}{2}, \frac{R_{max}}{4}, \frac{R_{max}}{8} \right\} \cap \{1, \dots, 2048\}. \quad (2)$$

In an NPDCCH region with R_{max} subframes, only devices requiring R as per (2) can be scheduled. Note that this restriction helps in reducing the blind decoding attempts by the NB-IoT device to four repetition values. In the absence of such restriction, the NB-IoT device must blindly search for all the configurations of $R \in \{1, \dots, 2048\}$. Further, the NB-IoT device requiring R repetitions can be scheduled only after every R th subframe location in R_{max} subframes. This constraint is explained with an example as follows. Consider an NPDCCH region with $R_{max} = 4$ subframes, and two NB-IoT devices requiring repetitions of $R = 1, 2$ subframes. Let the base station schedule the device with $R = 1$ in the first subframe. Now, the second NB-IoT device which requires $R = 2$, cannot be scheduled starting from the second subframe. As per the 3GPP constraints, the device with $R = 2$ can be scheduled only after every 2 subframes. Thus, the DCI for the second NB-IoT device is scheduled from the third subframe, as shown in Fig. 1. These constraints limit the number of blind decoding attempts that an NB-IoT device has to perform [19]. However, as explained later in Section 3.2, in case of improper NPDCCH scheduling, these constraints can result in a significant resource wastage and power consumption by the NB-IoT devices.

3.2. NPDSCH scheduling

The base station can schedule the data intended for the NB-IoT device in NPDSCH only after four subframes from the end of its corresponding DCI in NPDCCH. The NB-IoT devices are assumed to be capable of performing one blind decoding in 1 ms. As explained in [19], to perform the blind decoding in all possible positions, the NB-IoT device requires an additional 4 ms of time. Hence, a four subframe gap is allowed between the data in NPDCCH and NPDSCH. Further, for an NB-IoT device, apart from this four subframe gap, there can be an additional configurable delay (k_0) for some DCI types. For the example scenario considered earlier, the DCI for the two NB-IoT devices, their NPDCCH regions, and the valid NPDSCH regions are shown in Fig. 1. In case the scheduling of the devices is not carried out optimally, this 4 subframe gap for each device can cause significant under-utilization of the resources.

The base station has the information about number of subframes of data (N_{SF}) that an NB-IoT device requires, and the repetition (R, N_{rep}) with which it needs to transmit that (control, shared channel)

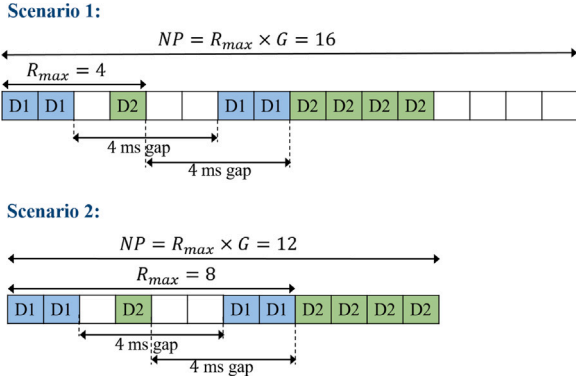


Fig. 2. Scheduling options for the two devices considered in the example.

information. The possible values of N_{SF} , N_{rep} , and a constant G as per 3GPP [3,20] are shown in (3)–(5).

$$N_{SF} \in \{1, 2, 3, 4, 5, 6, 8, 10\}. \quad (3)$$

$$N_{rep} \in \{1, 2, 4, \dots, 2048\}. \quad (4)$$

$$G = \{1.5, 2, 4, 8, \dots, 64\}. \quad (5)$$

The scheduler at the base station has to decide the R_{max} and G values based on the data and the repetition requirement by the NB-IoT devices. The base station then broadcasts NP as the region of subframes that the NB-IoT devices have to look for the NPDCCH and NPDSCH. The NP is formulated based on these two factors, as shown in (6).

$$NP = R_{max} \times G. \quad (6)$$

Readers unfamiliar with the NPDCCH and NPDSCH can refer [17, 18] for further details. Next, we explain the need for joint scheduling of NPDCCH and NPDSCH with an example, as follows.

Example

We consider two devices D1, D2 requiring DCI repetitions of $R = 2, 1$ subframes, and the shared channel data of $N_{SF} \times N_{rep} = 2, 4$ subframes, respectively. In Fig. 2, we present two of the various possible scheduling scenarios. Scenario 1 considers NPDCCH scheduling followed by NPDSCH scheduling, whereas, Scenario 2 considers joint scheduling of NPDCCH and NPDSCH. In Scenario 1, we first schedule the devices in NPDCCH within the least possible number of subframes. Considering the constraints (1)–(2), the least R_{max} value that can accommodate both the DCIs with $R = 2, 1$ is $R_{max} = 4$. When $R_{max} = 4$, to schedule the devices in NPDSCH, as per (5), the least possible value of G is 4. In Scenario 2, we consider both NPDCCH and NPDSCH resource requirements and schedule the devices with $R_{max} = 8, G = 1.5$ while satisfying all the 3GPP constraints (1)–(6). Note that the number of subframes not utilized is 7 in Scenario 1, and 3 in Scenario 2. Thus, the scheduling algorithm that simultaneously considers the joint resource allocation can reduce the unallocated subframes and comparatively pack more devices in a radio frame. These factors significantly impact resource utilization and device battery power; and are the key motivations for this work. Next, we present the proposed scheduling algorithm. We try to optimize the resource utilization and power consumption of the devices while performing joint scheduling of NPDCCH and NPDSCH in the proposed algorithm.

4. Proposed algorithms

In this section, we initially propose an adaptive resource allocation algorithm. We define the performance metrics for evaluating the NB-IoT scheduling algorithms and propose the joint scheduling algorithm. We also discuss the existing NB-IoT scheduling algorithms that we use for comparison. We then present a receiver design for the NB-IoT systems.

4.1. Adaptive repetition allocation algorithm

The base station has information about the operating signal to interference-plus-noise ratio (SINR) of each NB-IoT device. Motivated by the Modulation and Coding Scheme (MCS) procedure in LTE and the resource allocation scheme presented in [10], we propose a novel method to choose R and N_{rep} for control and shared channel, respectively. For the allocation of repetitions, we rely on the link-level simulations carried out for all possible repetition values over a wide-range of SINR values. The DCIs in NB-IoT are meant for either a particular NB-IoT device or a group of devices. The DCIs meant for individual NB-IoT devices are transmitted in UE-specific Search Space (USS) of the control channel region, whereas, the DCIs meant for a group of NB-IoT devices are transmitted in Common Search Space (CSS) of the control channel region. We initially define a reference SINR as follows. For the DCI transmitted in USS, the reference SINR is same as the operating SINR of the NB-IoT device, whereas, for the DCI transmitted in CSS, the reference SINR is the least SINR among all of the NB-IoT devices. We initially carry-out link-level simulations for each repetition value and tabulate the block error rate (BLER) for varying SINR values. Now, for each DCI with a reference SINR, we assign a repetition value R such that the BLER at the reference SINR is smaller than β [10]. Typically, β is decided by the industry operator based on the type of service (for example, $\beta = 0.1$ can be used for regular sensor data, whereas, $\beta = 0.001$ can be used for emergency/critical services, etc.). Similarly, for a payload of N_{SF} of data in the shared channel, we choose N_{rep} such that the BLER at the reference SINR is smaller than β . Note that the SINR of a device changes over time because of the varying channel propagation, interference characteristics, network scenarios, etc. This SINR (which includes the impact of channel model and network scenario) is used for mapping repetition-levels to any device. Hence, irrespective of the network scenario, the proposed algorithm ensures that the probability of failed transmissions over time converges to the chosen threshold (β). Thus, the link-performance (BLER) varies with the selected threshold, but the adaptive nature of the proposed mechanism is independent of the network scenario. After assigning repetition values R and N_{rep} for each NB-IoT device, the scheduler then decides the devices that are to be allocated with resources in each subframe. Next, we present the performance metrics and then propose the scheduler design.

4.2. Performance metrics

4.2.1. Resource utilization (χ)

As discussed in Section 3.2, there are possibly many ways to perform the scheduling (or deciding the NP). There can be resource wastage because of the various repetition requirements R , R_{max} , and G configurations, and the configurable delay between the NPDCCH and NPDSCH. A good scheduler should minimize the number of resources wasted. Hence, we consider resource utilization (χ) as a performance metric and define it as the ratio of the number of subframes allocated to the number of (#) subframes available in the NP.

$$\chi = \frac{\# \text{ subframes allocated to the NB-IoT devices}}{\# \text{ subframes in the NP}}.$$

A larger value of χ , closer to 1, indicates a better resource utilization. Note that the maximum value of χ is 1, where, all the subframes in an NP are allocated to the NB-IoT devices.

4.2.2. Number of devices scheduled per NP (η)

The NB-IoT devices do not know the exact location of the DCI in the NPDCCH. Hence, when the base station broadcasts the NP, all the active devices start decoding the NPDCCH region for their DCI. However, as per 3GPP specification [3], a maximum of eight devices can be scheduled in any NPDCCH. The number of scheduled devices in each NP is inversely proportional to the overall power consumption of the

NB-IoT devices under that base station. Hence, we consider the average number of devices scheduled per NP (η) as a performance metric.

η = Average number of devices scheduled per NP.

A larger value of η indicates the average power consumption of devices will be less, and thus, a better scheduling algorithm. The maximum permitted value of η as per the constraints in [3] is 8.

4.3. Proposed joint scheduler

4.3.1. Scheduler design and algorithm

We pick the first NB-IoT device in the order from the scheduling list and calculate all the possible configurations of forming an NP as per (1)–(6). For each configuration of NP, we try to fit the other NB-IoT devices by maximizing the formulated objective function (ζ_{iu} in (7)). We then consider the configuration which has the maximum value for objective function (ζ_{iu}) and schedule the NB-IoT devices. The scheduling procedure is explained in detail as follows.

1. We initially sort the NB-IoT devices in the order of scheduling (based on device priority, if any). We pick the first NB-IoT device in the sorted list and let the repetition required by that NB-IoT device for transmitting DCI in NPDCCH be R_i . As per 3GPP constraints presented in (1)–(2), the DCI for the device can be scheduled only in limited R_{max} and G values. For each possible configuration u , we calculate corresponding $R_{max_{i,u}}$ and $G_{i,u}$ values for the chosen R_i , and consider only these configurations while maximizing the objective function.
2. We formulate the objective function as a set-packing problem to achieve optimal resource utilization (χ) and maximize the number of devices scheduled per NP (η). For each possible configuration of $R_{max_{i,u}}$ mentioned in (1)–(2) and for each value of $G_{i,u}$ mentioned in (5), we fit the devices in the NP by maximizing the following objective function.

$$\zeta_{iu} = \max_{X_{iu}} [\alpha \widetilde{\eta}_{iu} + (1 - \alpha) \widetilde{\chi}_{iu}] X_{iu},$$

$$s.t. \quad A_{iu} X_{iu} \leq 1_r. \quad (7)$$

where, A_{iu} is the constraint matrix constructed to meet (1)–(6), X_{iu} is the scheduler output for the u th configuration of the i th NP, $\widetilde{\eta}_{iu}$ and $\widetilde{\chi}_{iu}$ are row vectors constructed to prioritize the devices based on the performance metrics η and χ for the configuration u of the i th NP, 1_r is a column vector of length r with all ones, and $\alpha (\geq 0)$ is the parameter to achieve various trade-offs between η and χ . With a smaller α , the objective function schedules the devices that maximize the resource utilization in the NP, and with a larger α , the objective function maximizes the number of devices in the NP. Thus, by tuning α , the network operator can achieve a trade-off between η and χ .

3. We calculate ζ_{iu} for all possible configurations. We then choose the configuration u , which has the maximum ζ_{iu} value and perform the scheduling of the devices in i th NP. Note that, along with $R_{max_{i,u}}$ and $G_{i,u}$, the scheduler also gives the optimal NB-IoT devices selection in X_{iu} .

Next, we explain the procedure for the implementation of the algorithm in detail.

4.3.2. Solving the objective function

The set-packing problem is a well-investigated topic, and there are a wide variety of efficient algorithms available in the literature to solve it [21–23]. Further, in [11], the authors have formulated and solved the scheduling in LTE as a set-packing problem. We follow a similar procedure, as mentioned in [11] for solving the objective function in this paper. We explain the formulation of the constraint matrix A_{iu} , and objective function ζ_{iu} with an example as follows.

Consider two devices with the first device requiring $R = 1$, $N_{rep} = 1$, $N_{SF} = 1$, and the second device requiring $R = 2$, $N_{rep} = 1$, $N_{SF} = 1$. The possible configurations for scheduling the first device as per (1)–(6) are $R_{max_i} = \{1, 2, 4, 8\}$, $G_i = \{1.5, 2, \dots, 64\}$. Consider the configuration $R_{max_{i,u}} = 4$ and $G_{i,u} = 2$. As per constraint (6), $NP = R_{max_{i,u}} \times G_{i,u} = 8$. We form a resource allocation matrix S_d for each device, with each row representing a subframe in the chosen NP, and each column representing possible ways of allocating resources for the NB-IoT device, as shown in (8).

$$\begin{array}{c} \text{R=1 has 4 possible} \\ \text{locations} \\ \begin{array}{c} \begin{array}{c} 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{array} \\ \begin{array}{c} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{array} \\ \begin{array}{c} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{array} \end{array} \\ \begin{array}{c} \updownarrow R_{max}=4 \\ \updownarrow NP=8 \end{array} \\ S_1 = \begin{array}{c} 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{array} \end{array} \quad , \quad \begin{array}{c} \text{R=2 has 2 possible} \\ \text{locations} \\ \begin{array}{c} 1 \\ 1 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{array} \\ \begin{array}{c} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{array} \end{array} \\ \begin{array}{c} \updownarrow NP=8 \end{array} \\ S_2 = \begin{array}{c} 1 \\ 1 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{array} \end{array} \quad (8)$$

When $NP = 8$, there will be at least one subframe not valid for NPDCCH and NPDSCH, as shown in the frame structure in Fig. 1. We consider a total of 9 subframes with 8 valid subframes, and hence, each S_d has 9 rows. In these 9 rows, the first 4 rows of each S_d represent the NPDCCH, 5th row represents an invalid subframe for NPDCCH/NPDSCH, and the last 4 rows represent the NPDSCH. For each possible way of scheduling the device, i.e., in each column of S_d in (8), 1 implies the subframe allocated for the device, 0 implies subframe not allocated for the device. Device 1 requires $R = 1$ subframe in NPDCCH, $N_{SF} \times N_{rep} = 1$ subframe in NPDSCH, and there are four such possible combinations within the NP. Device 2 requires $R = 2$ subframes in NPDCCH, and $N_{SF} \times N_{rep} = 1$ subframe in NPDSCH, and there are two such possible combinations within the NP. Note that there is a 4 ms gap between the data in NPDCCH and NPDSCH for each NB-IoT device.

The vector X_{iu} for the configuration $R_{max_{i,u}}$, $G_{i,u}$ is constructed as follows.

$$\begin{aligned} X_{iu} &= [A_{u,1} \ A_{u,2} \ \dots \ A_{u,M}]^T, \\ A_{u,m} &= [\mu_{u,m,1} \ \mu_{u,m,2} \ \dots \ \mu_{u,m,K_m}], \\ \mu_{u,m,j} &\in \{0, 1\}. \end{aligned} \quad (9)$$

where, M is the number of active devices, K_m is the number of possible locations for scheduling m th device in the given NP, and $\mu_{u,m,j} = 1$ represents that the j th location is chosen for scheduling the m th device in the configuration u . Note that X_{iu}^* is the resultant vector obtained by maximizing the objective function ζ_{iu} . Thus, based on (8)–(9), we formulate the resource constraints of allocating a subframe to only one device in the NP as follows.

$$\begin{aligned} B_{iu} X_{iu} &\leq 1_{NP}, \\ B_{iu} &= [S_1 \ S_2 \ \dots \ S_M], \end{aligned} \quad (10)$$

where, 1_{NP} is a column vector with all ones of length NP and S_i 's are the matrix for each device as discussed in (9). Further, we formulate the constraint of scheduling a device only once in the NP as follows,

$$\begin{aligned} C_{iu} X_{iu} &\leq 1_M, \\ C_{iu} &= \begin{bmatrix} 1_{K_1}^T & 0_{K_2}^T & \dots & 0_{K_M}^T \\ 0_{K_1}^T & 1_{K_2}^T & \dots & 0_{K_M}^T \\ \vdots & \ddots & \ddots & \vdots \\ 0_{K_1}^T & \dots & \dots & 1_{K_M}^T \end{bmatrix}. \end{aligned} \quad (11)$$

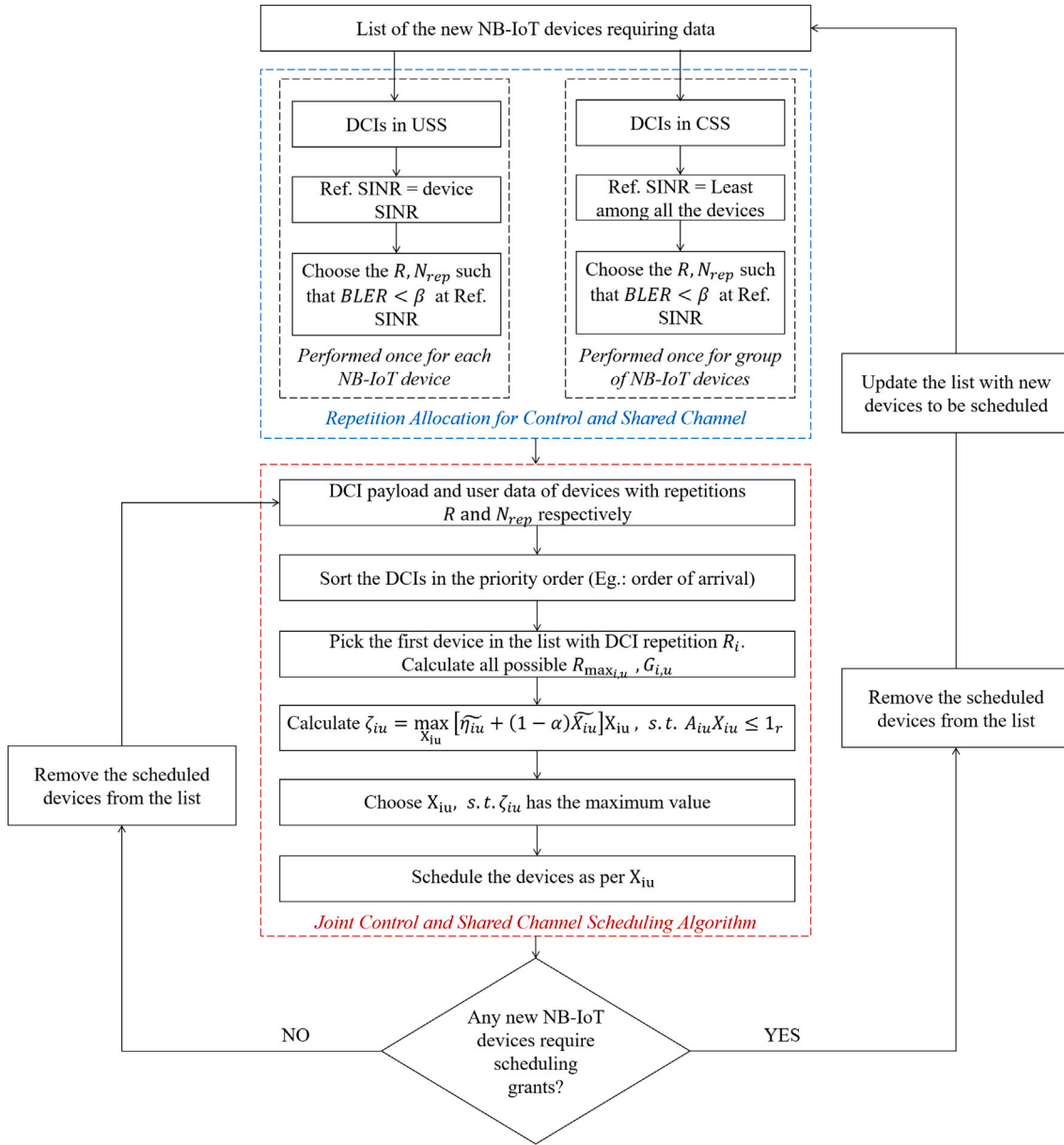


Fig. 3. End-to-end procedure of Repetition Allocation and Joint Scheduling Algorithm.

We then stack the constraints (10)–(11) into a single matrix A_{iu} as follows.

$$A_{iu} = \begin{bmatrix} S_1 & S_2 & \cdots & S_M \\ 1_{K_1}^T & 0_{K_2}^T & \cdots & 0_{K_M}^T \\ 0_{K_1}^T & 1_{K_2}^T & \cdots & 0_{K_M}^T \\ \vdots & \ddots & \ddots & \vdots \\ 0_{K_1}^T & \cdots & \cdots & 1_{K_M}^T \end{bmatrix}.$$

Thus, $A_{iu} X_{iu} \leq 1_r$, where, $r = NP + M$, captures all the 3GPP constraints for scheduling the NB-IoT devices in an NP. Next, we explain the formulation of the objective function.

We formulate η_{iu} as $\eta_{iu} = 1_r^T$ to maximize the number of devices scheduled in the NP. With $\eta_{iu} = 1_r^T$, maximizing $\eta_{iu} X_{iu}$ is the same as maximizing $\sum_{u,m,j} \mu_{u,m,j}$, which in turn maximizes the number of devices scheduled in the NP. We formulate χ_{iu} as $\chi_{iu} = [\sigma_1, \sigma_2, \dots, \sigma_r]$, where σ_j is the sum of elements in the j th column of B_{iu} matrix. Thus, maximizing $\chi_{iu} X_{iu}$ is the same as maximizing the number of resources utilized in an NP. However, the range of values is different in η_{iu} and χ_{iu} , and hence, we use the normalized $\tilde{\eta}_{iu}$ and $\tilde{\chi}_{iu}$ while maximizing

the objective function ζ_{iu} . A pictorial representation for the end-to-end procedure of resource allocation and scheduling is presented in Fig. 3. Next, we briefly discuss some of the NB-IoT scheduling algorithms from the existing literature that we have implemented for the numerical comparison and performance evaluation of the proposed scheduler.

4.4. State-of-the-art NB-IoT scheduling algorithms

In [19], authors follow a sorting procedure and schedule the NB-IoT devices in the control channel. We have implemented the maximum repetition (Max- R_i), minimum repetition (Min- R_i), and baseline schedulers proposed in [19] for the comparison.

4.4.1. Max- R_i scheduler

The scheduler sorts the devices in the descending order of their repetition requirements R_i . This sorting helps in stacking the devices that fit in an R_{max} side-by-side. Then, the scheduler picks devices from the sorted list and fits them in a suitable R_{max} by minimizing the resource wastage. Note that this scheduling procedure maximizes only

the NPDCCH resource utilization. The size of NP is later decided based on the chosen R_{max} . Further, the algorithm ignores the priority or order of scheduling the devices as it schedules the devices based on the sorted list.

4.4.2. Min- R_i scheduler

Except for sorting the devices in the ascending order, the scheduling procedure is similar to Max- R_i scheduler.

4.4.3. Baseline scheduler

In the baseline scheduler, the base station picks the devices in the priority order from the list of devices. The scheduler checks for the R_{max} and G values which give minimum NP such that the NP can accommodate R subframes in NPDCCH and N_{SF} subframes of the data with N_{rep} repetitions in NPDSCH.

4.5. Analysis of the scheduling algorithms

In this section, we derive the upper and lower bounds on the performance gains with the proposed scheduler. In Lemma 1, we prove that the performance of the proposed scheduler in terms of resource utilization with $\alpha = 0$ is same or better than the remaining schedulers. Further, we quantify the maximum performance gain we achieve with the proposed scheduler. Note that a larger value of χ is better and corresponds to a better resource utilization.

Lemma 1. Let $\chi_{algorithm}$ denote the resource utilization of a scheduling algorithm. Then, for any configuration of NB-IoT device distribution, the following holds:

$$\chi_{Remaining_Schedulers} \leq \chi_{Proposed|\alpha=0} \leq \frac{64}{34} \chi_{Remaining_Schedulers}$$

Proof. For any R , consider the configuration $C = \{R, 8R, 64R, 512R\}$, where each element of the C represents the resource requirement by an NB-IoT device for transmitting a DCI. This resource requirement configuration forms the worst-case scenario for the proposed scheduler. With this configuration, as per the constraint (2), no two devices can be grouped together into a single search space by any scheduler, and hence, all the schedulers have the same resource utilization. Thus,

$$\chi_{Proposed|\alpha=0}^{worst} = \chi_{Remaining_Schedulers}$$

For any R , consider the configuration $C = \{\frac{R}{8}, \frac{R}{2}\}$ and $S = \{30R, R\}$, where each element of the C and S represents the resource requirement by an NB-IoT device for transmitting a DCI and a transport block, respectively. This resource requirement configuration forms the best-case scenario for the proposed scheduler. With this configuration, all the existing schedulers initially try to group the two devices in a search space of length $R_{max} = R$. Then, the least possible G that could fit the resources is $G = 64$. However, the proposed scheduler with $\alpha = 0$, schedules the devices separately in two search spaces $R_{max_1} = \frac{R}{8}, G_1 = 16$, and $R_{max_2} = \frac{R}{2}, G_2 = 64$. Note that the total resources used by the other schedulers is $64R$, whereas, the proposed scheduler with $\alpha = 0$ uses $2R + 32R = 34R$. Thus,

$$\chi_{Proposed|\alpha=0}^{best} = \frac{64}{34} \chi_{Remaining_Schedulers}$$

This completes the proof of Lemma 1. $\square$

As shown in Lemma 1, the proposed scheduler has better resource utilization when compared to the remaining schedulers. Thus, it can accommodate more number of NB-IoT devices when compared to other schedulers, and thus, achieve better system capacity.

In Lemma 2, we quantify the performance of the proposed scheduler in terms of η . Note that a larger value of η is better and corresponds to lower power consumption by the devices.

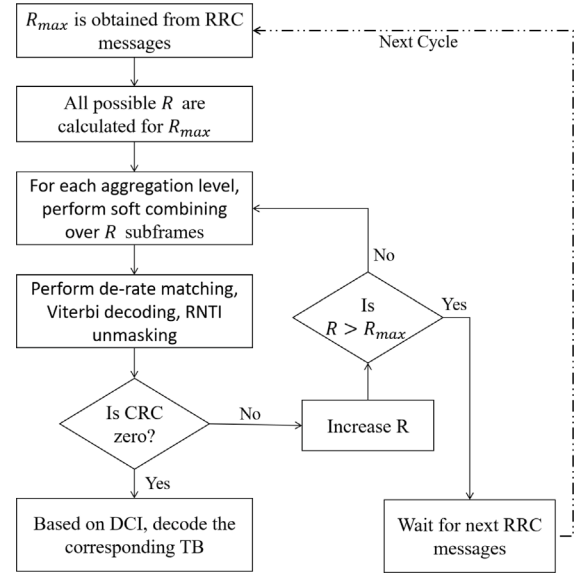


Fig. 4. Blind Decodings in NB-IoT.

Lemma 2. Let $\eta_{algorithm}$ denote the number of devices scheduled per NP by a scheduling algorithm. Then, for any configuration of NB-IoT device distribution, the following holds:

$$\eta_{Remaining_Schedulers} \leq \eta_{Proposed|\alpha=1} \leq 4\eta_{Remaining_Schedulers}$$

Proof. For any R , consider the configuration $C = \{R, 8R, 64R, 512R\}$, where each element of the C represents the resource requirement by an NB-IoT device for transmitting a DCI. This resource requirement configuration forms the worst-case scenario for the proposed scheduler. With this configuration, as per the constraint (2), no two devices can be grouped together into a single search space by any scheduler, and hence, all the schedulers have the same number of devices per NP. Thus,

$$\eta_{Proposed|\alpha=1}^{worst} = \eta_{Remaining_Schedulers}$$

For any R , consider the configuration $C = \{\frac{R}{8}, \frac{R}{8}, \frac{R}{8}, \frac{R}{8}\}$ and $S = \{8R, 8R, 8R, 8R\}$, where each element of the C and S represents the resource requirement by an NB-IoT device for transmitting a DCI and a transport block, respectively. This resource requirement configuration forms the best-case scenario for the proposed scheduler. With this configuration, all the existing schedulers initially try to group the four devices in a search space of length $R_{max} = \frac{R}{2}$. However, no choice of G could fit all the resources, and hence, they drop users and try for a feasible G . They end up scheduling the devices individually in separate search spaces. The proposed scheduler with $\alpha = 1$, ingeniously picks $R_{max} = R, G = 64$, which fits all the devices' resource requirements. Thus,

$$\eta_{Proposed|\alpha=1}^{best} = 4\eta_{Remaining_Schedulers}$$

This completes the proof of Lemma 2. $\square$

As shown in Lemma 2, the proposed scheduler with $\alpha = 1$ accommodates more number of devices per NP, when compared to the remaining schedulers, and thus, results in lower power consumption of the NB-IoT devices. Next, we present the adaptive repetition allocation algorithm.

4.6. Receiver design for NB-IoT

In NB-IoT system, the device does not know the repetition and the location of the DCI in a search space. Hence, the device has to consider

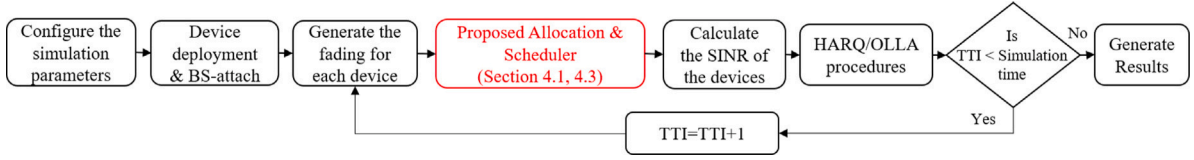


Fig. 5. Block diagram of system-level simulation with proposed algorithms.

Table 2
Simulation Parameters.

Parameter	Value
Bandwidth	180 KHz (1 PRB)
BS transmit Power	44 dBm
Carrier frequency	900 MHz
Cellular Layout	Hexagonal Grid, 7 sites, 3 sectors per site
G	{1.5, 2, ..., 64}
Inter-site distance	1732 m
NB-IoT device location	1000 devices dropped uniformly in entire cell
NB-IoT Mode of Operation	Standalone (used in link-abstractions)
N_{rep}	{1, 2, ..., 2048}
N_{SF}	{1, 2, ..., 10}
Pathloss model	$L=120.9 + 37.6\log_{10}(R)$
R_{max}	{1, 2, ..., 2048}
UE transmit Power	23 dBm

all the possible configurations and perform blind decoding for DCI. In this section, we present a receiver design and explain the blind decodings of NB-IoT in detail.

Initially, the device will obtain the search space length of the control channel, R_{max} from RRC messages. Then, the device calculates all possible configurations of R , as mentioned in (2). For each configuration, the device performs demodulation and descrambling of R subframes, and soft combines the data present in these R subframes. The device then performs the other physical layer procedures like de-rate matching, Viterbi decoding, and CRC check. If the CRC results in a zero-remainder, then the device assumes the DCI to be valid. The DCI carries the scheduling information about the transport block transmitted in the shared channel. Thus, the NB-IoT device obtains the scheduling related information from DCI and decodes the actual data transmitted in the shared channel. In case the CRC is failed, then it increases the repetition value R and performs the decoding procedure for all possible configurations again. This procedure is repeated until all possible configurations of R are exhausted or a successful CRC check. If CRC is failed for all the configurations, then either the DCI for the device was not transmitted in the search space or the repetition level transmitted was not sufficient for the device to decode the data. In both cases, the NB-IoT device has to wait for the next RRC messages. A pictorial representation of the procedure is presented in Fig. 4. Next, we present the simulation results.

5. Results and discussion

5.1. System level simulations

A pictorial representation of the simulation procedure is presented in Fig. 5. We initially configure the simulation parameters [24], as mentioned in Table 2. We then deploy the NB-IoT devices randomly in the sector and perform a device-attach to the base station based on the received signal power. We generate the channel between each NB-IoT device and base station link for every subframe. We implement the proposed repetition allocation scheme presented in Section 4.1 to assign repetitions to each device. Based on the SINR of each NB-IoT device, the repetition for control channel R and shared channel N_{rep} are selected assuming BLER threshold of 0.01. Further, for each NB-IoT device, we chose the NPDSCH data N_{SF} randomly from the set {1, 2, ..., 10}. We then perform the scheduling of the devices by solving

the objective function presented in Section 4.3. The proposed scheduler picks the first device in the order and calculates all possible $R_{max_{i,u}}$, and $G_{i,u}$ values. The scheduler computes the objective function ζ_{iu} for all these configurations. Then, the scheduler chooses the $R_{max_{i,u}}$ and $G_{i,u}$ configuration which has the largest ζ_{iu} value, and schedules the devices. To solve the objective function, we use the MATLAB built-in function *intlinprog*. We then calculate the SINR and perform the appropriate Hybrid ARQ and outer-loop link adaptation (OLLA) procedures for the scheduled devices. We continue these procedures for a pre-configured simulation time and evaluate the performance metrics. The proposed joint scheduling algorithm is run for $\alpha = 0.1, 0.5, 0.9$. For the same set of devices, we run the Max- R_i , Min- R_i , and the baseline scheduler. We calculate the mean number of devices scheduled in each NP ($\bar{\eta}$) and the average resource utilization ($\bar{\chi}$) for each of these scheduling algorithms.

In Fig. 6, the cumulative distribution function (CDF) of ($\bar{\chi}$) is presented, and in Fig. 7 the mean of number of devices scheduled in an NP ($\bar{\eta}$) is plotted. With larger α , the objective function (ζ_{iu}) tries to maximize the number of devices scheduled in each NP. Thus, the proposed scheduler with $\alpha = 0.9$ has a larger number of devices scheduled in Fig. 7. However, the scheduler does not prioritize to minimize resource wastage and aggressively schedules more devices. Hence, the scheduler with $\alpha = 0.9$ has poor resource utilization as shown in Fig. 6. With decreasing α , the resource utilization ($\bar{\chi}$) is maximized as shown in Fig. 6, and the number of devices scheduled in an NP ($\bar{\eta}$) is minimized as shown in Fig. 7.

The baseline scheduler always schedules a single device, and hence, $\eta=1$. The Max- R_i and Min- R_i algorithms sort the devices and schedule more number of devices compared to baseline scheduler. Even though the algorithms maximize the NPDCCH resource utilization, the schedulers cannot fit the data for all these devices in the available NP sizes. Hence, these schedulers give minimal gains in terms of number of devices scheduled per NP. The average resource utilization ($\bar{\chi}$) of the Max- R_i , Min- R_i , and the baseline scheduler are significantly lower than the proposed scheduler for various values of α . The median of resource utilization ($\bar{\chi}$) is 0.6 in the Max- R_i scheduler, and is close to 0.9 with the proposed scheduler (for $\alpha = 0.1$). Thus, with the proposed scheduler, 50% additional resources can be utilized, or three times the number of scheduled devices (η) can be achieved with suitable selection of α .

In Table 3, a numerical analysis of all the schedulers across various performance metrics is presented. The schedulers having the best performance metrics are highlighted in the table for a better understanding. The proposed scheduler with $\alpha = 0.1$ takes the least time to schedule all the devices and has the least number of physical resource blocks unutilized. The proposed scheduler with various α outperforms all the existing schedulers. Note that the Max- R_i scheduler has the best performance among all the NB-IoT schedulers from the literature, and our proposed scheduler with $\alpha = 0.1$ performs five times better in terms of resource utilization when compared to the Max- R_i scheduler.

The number of NPs broadcast is directly proportional to the power consumption of the devices. The baseline scheduler broadcasts one NP for every device and thus, has 1000 NPs broadcast. The Max- R_i and Min- R_i algorithms comparatively schedule more number of devices in an NP, and hence, take less number of NPs to schedule all the devices. The proposed scheduler with various α outperforms all the existing schedulers. The proposed scheduler with $\alpha = 0.9$ takes one-third of the NPs compared to the Max- R_i scheduler. Further, it schedules three

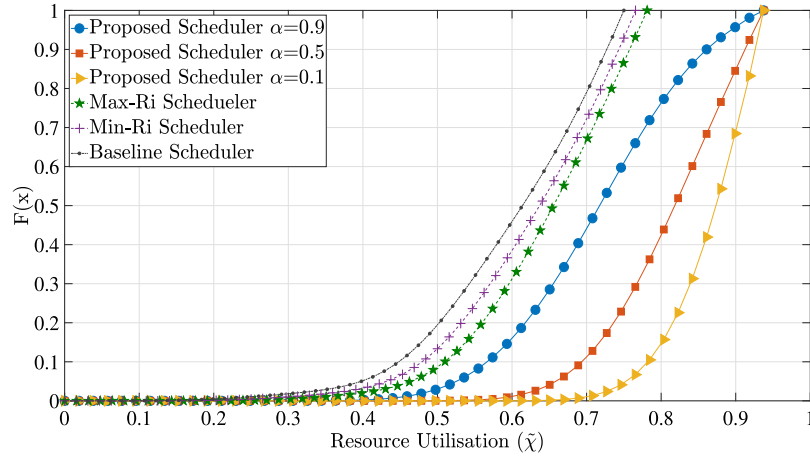


Fig. 6. CDF of resource utilization ($\hat{\chi}$) for various scheduling algorithms.

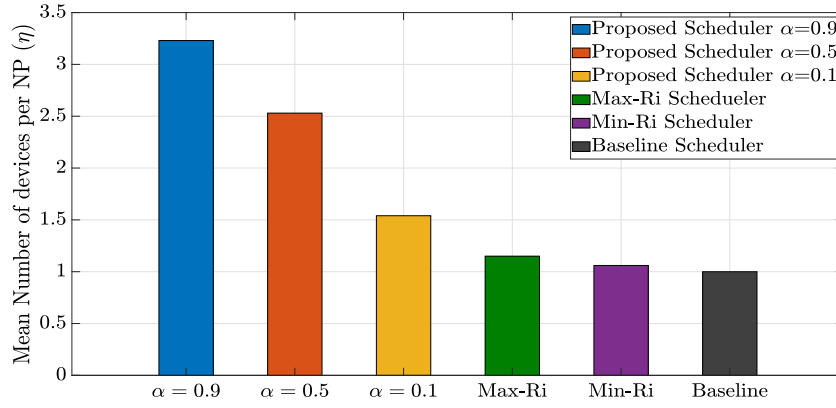


Fig. 7. Histogram of mean number of devices per NP for various scheduling algorithms.

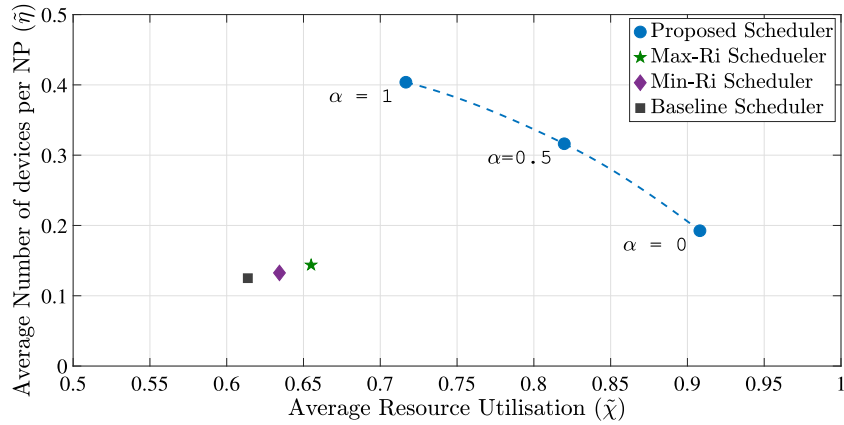


Fig. 8. Trade-off between average resource utilization ($\hat{\chi}$) and average power consumption ($\bar{\eta}$) for various scheduling algorithms.

times more the number of devices in an NP when compared to other schedulers. Note that the main reason for the poor performance of the existing NB-IoT schedulers is that they do not consider the resources in both NPDCCH and NPDSCH simultaneously.

In Fig. 8, we present the trade-off between resource utilization and the number of scheduled devices. The proposed scheduler performs significantly better than all the existing scheduling algorithms. Further, the proposed scheduler allows the network operator to choose the α to achieve various trade-offs between resource utilization and the number of devices scheduled.

5.2. Exhaustive link simulation and analysis

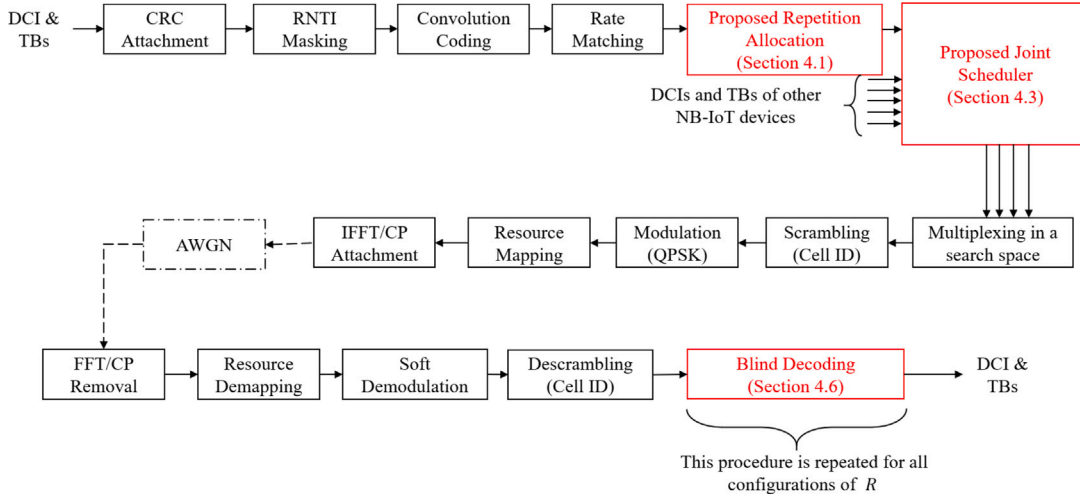
We have implemented the whole transmitter and receiver algorithms of control and shared channels of NB-IoT as per the 3GPP specifications [1–3]. We have then integrated the proposed repetition allocation algorithm, along with the proposed joint scheduling algorithm and an exhaustive link simulation, as shown in Fig. 9 is carried out as follows.

We consider 1000 users for the simulation. We then assign an arbitrary SINR to each user by randomly picking from a uniform distribution ranging from -12 to 15 dB. For each user, the DCI and

Table 3

Detailed numerical analysis of various performance metrics for all the scheduling algorithms.

Scheduler type	Time taken by the base station to schedule all the devices		Total number of unutilized PRBs	Number of NPs broadcast by the base station	Mean number of devices per NP
	Subframes	Minutes			
Proposed Scheduler $\alpha = 0.9$	2451.1×10^3	40.851	740.72×10^3	328	3.23
Proposed Scheduler $\alpha = 0.5$	2166.6×10^3	36.11	456.28×10^3	432	2.53
Proposed Scheduler $\alpha = 0.1$	1884.2×10^3	31.4	173.65×10^3	800	1.54
Max- R_i Scheduler	2652.8×10^3	44.214	942.49×10^3	890	1.15
Min- R_i Scheduler	2734.8×10^3	45.58	1024.5×10^3	970	1.06
Baseline Scheduler	2823.7×10^3	47.061	1113.3×10^3	1000	1

**Fig. 9.** Block diagram of exhaustive link simulation with transmitter and receiver chain along with proposed algorithms.

the transport block (TB) undergo the physical layer processing like cyclic redundancy attachments, convolution coding, rate matching, etc., as per the 3GPP specifications [1–3]. The resultant data is fed to the proposed repetition allocation algorithm, which decides the repetition required for each payload, as mentioned in Section 4.1. All these payloads with various repetition requirements are then fed to the proposed joint control and shared channel scheduling algorithm. The algorithm selects the users to be multiplexed in a search space, as mentioned in 4.3. The multiplexed data undergoes further physical layer processing like scrambling, modulation, and is then mapped to the resource elements of a resource grid. In the end, the resource grid is converted to set of OFDM symbols by using IFFT and a cyclic prefix (CP) attachment. The noise is added to each OFDM symbol based on the SINR of each user scheduled in that symbol. At the receiver, the corresponding signal processing is carried out, which involves, FFT operation, CP removal, resource de-mapping, soft demodulation, and descrambling. However, the receiver does not know the repetition used for the DCI. Hence, it carries out blind decoding for all possible repetition sizes, which involves de-rate matching, Viterbi decoding, and cyclic redundancy check (CRC). If the CRC is passed, we consider the payload to be successfully decoded.

We have performed the above simulation procedure for various scheduling algorithms discussed earlier, and for each scheduling algorithm, we have calculated the resource utilization (χ), the mean number of devices per NP (η), and the block error rate. Note that these metrics evaluate the performance of the proposed joint scheduler and the repetition allocation algorithm. A summary of the results has been presented in Table 4. Since the resource allocation algorithm considers 1% BLER as a reference for allocation, all the scheduling algorithms have close to 1% BLER. Further, the proposed scheduler with any α has a better χ and η when compared with the state-of-the-art algorithms.

Table 4

Summary of results with exhaustive link-level simulations.

Scheduler Type	χ	η	BLER
Proposed Scheduler, $\alpha = 0.9$	0.65	2.7	1.5%
Proposed Scheduler, $\alpha = 0.5$	0.71	1.8	1.1%
Proposed Scheduler, $\alpha = 0.1$	0.82	1.4	1.2%
Max- R_i Scheduler	0.58	1.2	1.6%
Min- R_i Scheduler	0.52	1.1	0.9%
Baseline Scheduler	0.51	1	0.9%

With $\alpha = 0.9$, the proposed scheduler has better η and with $\alpha = 0.1$, it has better χ .

6. Conclusion and future work

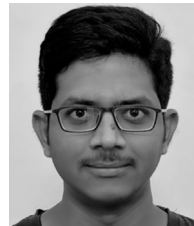
In this paper, we have explained the rationale for jointly scheduling the NB-IoT devices in both control and shared channel. We have formulated the resource allocation as a set packing problem and proposed a joint scheduling algorithm. We have also proposed an adaptive repetition allocation algorithm and presented a receiver design for NB-IoT. Further, we have defined performance metrics to evaluate the scheduling algorithms in NB-IoT and presented the performance bounds of the proposed scheduler. We have evaluated and compared the performance of the proposed scheduler using system-level and link-level simulations. The simulation results have shown that the proposed scheduler significantly outperforms the current state-of-the-art algorithms. With the proposed scheduler, the network operators can choose a suitable trade-off between resource utilization and power consumption of the NB-IoT devices. In future, we will implement the proposed scheduler for an NB-IoT system and evaluate the performance on hardware testbeds.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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